Theory of Computation and automata

Assignment 2

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Problem 1. (Regular Expressions-NFA-DFA) Write a corresponding regular expression for the following languages then convert to NFA->DFA. where $\sum = \{a, b\}$.

a) $L = \{abwa: w \in \{a, b\}^*\}$
b) $L = \{w_1 abw_2 : w_1 \in \{a, b\}^* \; and \; w_2 \in \{a, b\}^*\}$
c) $L = \{w: |w| mod \; 4 = 0\}$

Problem 2. (Regular Expressions-NFA-DFA) Consider the set of strings on $\sum = \{0, 1\}$ defined by the requirements below. Find regular expression and NFA->DFA for the following:

a) All strings containing 00 but not 000,
b) The leftmost symbol differs from the rightmost one.
c) All strings in which the leftmost two symbols and the rightmost two symbols are identical.
d) Every 00 is followed immediately by a 1, for example, the strings 1001001, 0010, 0010011001 are in the language, but 0001, 000, 00 and 00010 are not.

Problem 3. (Properties of Regular Languages)

a) The symmetric difference between languages L_1 and L_2, denoted $\mathbf{L_1 \ominus L_2}$ consists of all the strings in the union of the two languages except the strings in the intersection. Show that if L_1 and L_2 are regular languages, then $\mathbf{L_1 \ominus L_2}$ is a regular language too.
b) For any language L let $\mathbf{L_{odd}}$ denote the language which consists of all strings in L that have odd length. Prove that if L is regular then $\mathbf{L_{odd}}$ is regular too.

Problem 4. (Regular Expressions) Show that the following languages, defined over the alphabet $\sum = \{0, 1\}$, are regular by giving the regular expressions that describe them:

a) L1= {all strings containing at least three 1}.
   Test strings: lambda, 100100, 0101011, 001110011000
b) L2 = {all strings with even length}.
   Test strings: lambda, 10, 110, 00000, 1110101110.
c) L3 = {all strings which do not contain substring 1100}.
   Test strings: lambda, 1001, 0011000, 10011010, 001100110.

NOTE: Some Problems require the software simulation tool for automata JFLAP (http://www.jflap,org/). The answers are the printouts of JFLAP. Use the Regular Expression option to write your regular expression. Since JFLAP does not support testing of strings on regular expressions, you will test your strings as follows. Use the Convert option to convert your regular expression to an NFA (by selecting the Do All and then the Export buttons). Run the tests cases on the NFA using the Multiple Run option, and print the result. So, for each language. you need to give the printout of the regular expression, and the multiple run on the corresponding NFA.